**Birla Institute of Technology & Science, Pilani**
**Work Integrated Learning Programmes Division**
**Second Semester 2025-2026**

**Mid-Semester Test**
**(EC-2 Regular)**

| | |
|---|---|
| Course No. | : AIMLZ565 |
| Course Title | : Machine Learning |
| Nature of Exam | : Closed Book |
| Weightage | : 30% |
| Duration | : 2 Hours |
| Date of Exam | : 15/03/2026 (FN/AN) |

No. of Pages = 4
No. of Questions = 6

Note to Students:

1. Please follow all the *Instructions to Candidates* given on the cover page of the answer book.
2. All parts of a question should be answered consecutively. Each answer should start from a fresh page.
3. Assumptions made if any, should be stated clearly at the beginning of your answer.

**Please Note:**
A **maximum of 12 main questions** with Q.Nos. 1 to N**. Please do not include any objective type questions. Please do not include any reference material like data tables in the question paper, as all Mid-Semester Test are completely Closed Book examinations.** Both the options to submit answers i.e. typing as well as scanning & uploading hand written answers should be available for ALL the questions.

Q.1 A city government wants to understand how annual sales tax collections change with the number of new car registrations each year. To support budgeting and revenue forecasting, the government collected the following historical data for 7 years, where:
X = Number of new car registrations (in thousands)

Y = Annual sales tax collections (in suitable units, e.g., million/crore)

| Year | X (Registrations in thousands) | Y (Sales tax collections) |
|---|---|---|
| 1 | 10 | 1 |
| 2 | 12 | 1.4 |
| 3 | 15 | 1.9 |
| 4 | 16 | 2 |
| 5 | 14 | 1.8 |
| 6 | 17 | 2.1 |
| 7 | 20 | 2.3 |

Assume a linear relationship between registrations and sales tax collection

$\hat{Y} = \theta_0 + \theta_1 X$ where the left hand side refers to the predicted value of Y.

(A) **[4 Marks]** Using the closed-form (Normal Equation) method of Ordinary Least Squares (OLS), determine the least-squares regression equation.

**Solution :**

One could either use the formulas to solve for the slope and the intercept or the closed form expression using the pseudoinverse if using the matrix vector notation.

The closed form expressions for the slope (b1) and intercept (b0) are :

$$b_1 = \frac{n\sum xy - (\sum x)(\sum y)}{n\sum x^2 - (\sum x)^2}$$

and

$$b_0 = \bar{y} - b_1\bar{x}$$

Where $\bar{x}$ and $\bar{y}$ are the sample means of x and y respectively

If using matrix vector notation, the overdetermined system can be expressed as :

$A\,\boldsymbol{\theta} = y$ and the parameters of the best fit model are given by

$\boldsymbol{\theta}^* = (A^T A)^{-1} A^T y$

Solving either way, the parameters of the best fit line are :

$\boldsymbol{\theta}_0 = -0.1581$ and $\boldsymbol{\theta}_1 = 0.1308$

Need to clearly show working and relevant intermediate steps to get full credit.

(B) **[2 Marks]** Using the fitted regression equation, estimate the annual sales tax collections when the city records 22,000 new car registrations (i.e., X=22).

**Solution :** Substituting X = 22 in the regression equation with parameters computed above yields, $\widehat{Y} = -.1581 + 0.1308 * 22 = 2.7203$ = 2.7 to one decimal place.

*(C)* **[4 Marks]** Consider the following training set with 5 examples and regression model as $Y=3-4X+2X^2$

| X | Y |
|---|---|
| 5 | 30 |
| 8 | 90 |
| 12 | 250 |
| 15 | 498 |
| 20 | 900 |

Calculate the following error measures.

- Root Mean Square Error
- Mean Absolute Error

**Solution :** The intermediate results are shown in the table below.

| X | Y | $\widehat{Y} = 3 - 4X + 2X^2$ | Absolute error | Squared error |
|---|---|---|---|---|
| 5 | 30 | 33 | 3 | 9 |
| 8 | 90 | 99 | 9 | 81 |
| 12 | 250 | 243 | 7 | 49 |
| 15 | 498 | 393 | 105 | 11025 |
| 20 | 900 | 723 | 177 | 31329 |

The Root Mean Squared Error = √(Mean of Squared error) = 92.188
Mean Absolute Error = 301/5 =60.2

Q.2

(a) **[4 Marks]** A dataset contains the following attributes: Temperature (°C), Salary (₹), Customer Satisfaction (Low/Medium/High), Product Category (Electronics, Grocery). Classify each as Nominal, Ordinal, Interval, or Ratio and justify.

**Solution :** Temperature (°C) and Salary are numerical data types while the others are categorical. The specific types are specified below. 1 mark for specifying the correct type for each attribute.

Temperature (°C) : Interval (0.5 marks each if the student has specified as numerical)
Salary (₹) : Ratio (0.5 marks each if the student has specified as numerical)
Customer Satisfaction : Ordinal, can also be argued to be nominal
Product Category : Nominal

(b) **[3 Marks]** A dataset records education level: High School → Bachelor's → Master's → PhD. What would be an accurate characterization of the data type for education level and explain which arithmetic operations (if any) such as computing the mean, are meaningful and which are not.

**Solution :** As there is a sequential ordering, it is fair to characterize the education level as ordinal. Comparison operations (< >) are meaningful. As it is not a numerical data type, differences and ratios are not meaningful. Computing the mean is not meaningful, however computing the median is meaningful.

(c) **[3 Marks]** Consider a dataset where the values of a numerical attribute for a set of instances are :

10, 12, 13, NaN, 15, 18, NaN, 20 where NaN denotes missing values.
Impute the missing values using mean and median imputation strategies.

**Solution :** Fill in all the missing values using the mean or median as specified.

Mean imputation : 14.667

Median imputation : 14

Q3

(a) **[5 Marks]** Consider a function **$J(\theta) = (\theta - 6)^2$** which we wish to minimize using gradient descent. Let the initial value for $\theta$ be 10, and let the learning rate $\alpha = 0.1$ (assume a fixed learning rate).
Perform two iterations of gradient descent and compute updated $\theta$ values.

**Solution :**
Gradient of the function **$J(\theta) = (\theta - 6)^2$** = grad(**$J(\theta)$**) = **$2*(\theta - 6)$**

Gradient descent update equation :

$$\theta^{(i+1)} = \theta^{(i)} - \alpha \text{ grad}(J(\theta)) = \theta^{(i)} - \alpha(2*(\theta-6))$$

The computations for the first two iterations are shown in the table below.

| **Iteration** | **θ** | **grad(J(θ))** |
|---|---|---|
| 0 | 10 | 2*(10-6)=8 |
| 1 | 10-0.1*8 = 9.2 | 2*(9.2-6)=6.4 |
| 2 | 9.2-0.1*6.4 =8.56 | |

(b) **[5 Marks]** A 3-class classifier produces the following confusion matrix:

| **Actual \ Predicted** | **A** | **B** | **C** |
|---|---|---|---|
| **A** | 40 | 5 | 5 |
| **B** | 8 | 30 | 12 |
| **C** | 3 | 7 | 35 |

Compute the following classification measures.

i) Accuracy

ii) Precision, Recall, and F1-score for each class

**Solution :**
i) Accuracy = Number of correctly classified instances / Total number of instances
= (40+30+35) / (40+5+5+8+30+12+3+7+35) = 0.7241 (72.41%)

ii) Precision, Recall, and F1-score for each class (F1 score = 2*Precision*Recall/(Precision + Recall)

| Class | Precision | Recall | F1Score |
|---|---|---|---|
| **A** | 0.7843 | 0.8 | 0.7921 |
| **B** | 0.7143 | 0.6 | 0.6522 |
| **C** | 0.6731 | 0.7778 | 0.7217 |

Q4

a. **[2 Marks]** Why can't we use the mean square error cost function used in linear regression for logistic regression?

**Solution :** A main reason for not using Mean Squared Error in Logistic regression is because it makes the cost function non-convex, resulting in getting trapped in local minima. Cross entropy (also called log-loss) which is often used in Logistic Regression, on the other hand results in a convex cost function. Another reason is that logistic regression is a classification method. Although an intermediate step involves regression, the output can be interpreted as a probability that the given instance belongs to the target class and hence is a number between 0 and 1. This is better expressed using a cross entropy loss rather than mean squared error.

b. **[3 Marks]** Consider the following dataset and explain with justification whether logistic regression can be used for perfect separation of the two classes.

| F1 | F2 | Class |
|---|---|---|
| -1 | -1 | positive |
| 1 | -1 | negative |
| -1 | 1 | negative |
| 1 | 1 | positive |

**Solution :** A simple plot of the above data shows that the classes are not linearly separable. As logistic regression can only compute linear decision boundaries, it cannot be used to achieve perfect separation in the above case.

c. **[5 Marks]** Can you use logistic regression for classification between more than two classes? How many binary classifiers would you need to implement one-vs-one for four classes? Explain how an instance is classified using multiple binary classifiers in the one-vs-one strategy.

**Solution :** Logistic regression is primarily a binary classification method, although it could be extended to multiple classes. A common method to apply binary classifiers to multi class classification is the one-vs-one strategy. Here binary classifiers are designed to discriminate between each pair of classes. An instance is assigned a class label corresponding to the highest number of “votes” (or highest aggregate confidence) obtained by the instance. For 4 classes, we need ${}^4C_2 = 6$ binary classifiers.

Q5

a. **[5 Marks]** How does L1 and L2 regularization affect bias and variance?

**Solution :** Regularization is used to control model complexity and prevent overfitting, in machine learning it is generally achieved by adding a penalty term to the loss function.
L1 regularization adds a penalty term = $\lambda\sum|w_i|$, this regularization is employed in Lasso for example.
L1 regularization typically increases Bias and decreases Variance.
L2 Regularization adds a penalty term = $\lambda\sum w_i^2$ L2 regularization is used in ridge regression for example, typically L2 regularization slightly increases Bias and reduces Variance

b. **[5 Marks]** Suppose the true function is: $y=x^3+\epsilon$ where $\epsilon \sim N(0,\sigma^2)$.

Three models are fitted:

Model 1: Linear regression

Model 2: Cubic regression

Model 3: 15th-degree polynomial

Answer the following questions and justify your answer in each case:

(i) Rank the models in terms of bias (highest to lowest).
(ii)Rank the models in terms of variance (highest to lowest).
(iii) Which model is expected to have the lowest test error? Explain using bias–variance decomposition.

**Solution :**
(i) Bias measures how far the model’s average prediction is from the true function.

- **Model 1 (Linear)**
  - Cannot represent x^3.
  - Very poor approximation.
  - **Highest Bias**
- **Model 2 (Cubic)**
  - Correct functional form.
  - Can perfectly represent x^3 (ignoring noise).
  - **Very Low Bias**
- **Model 3 (Degree 15)**
  - Very flexible.
  - Can represent cubic function easily.
  - **Low Bias (similar to Model 2)**

**Bias Ranking:** Model 1>Model 3≈Model 2

**(ii)** Variance measures sensitivity to training data fluctuations.

- **Model 3 (Degree 15)**
  - Extremely flexible.
  - Highly sensitive to small data changes.
  - **Highest Variance**
- **Model 2 (Cubic)**
  - Moderate flexibility.
  - Stable but still data-dependent.
  - **Moderate Variance**
- **Model 1 (Linear)**
  - Very simple.
  - Predictions change little with different samples.
  - **Lowest Variance**

**Variance Ranking:**

Model 3>Model 2>Model 1

(iii) **Model 2 (Cubic regression) is expected to have the lowest test error** as it matches the true functional form exactly, minimizing bias without excessive variance.

Q6

Consider the following training example dataset, where a1,a2 and a3 refer to attributes and Class refers to the class label (Positive, Negative).

| **InstanceID** | **a1** | **a2** | **a3** | **Class** |
|---|---|---|---|---|
| 1 | T | T | 1 | + |
| 2 | T | T | 6 | + |
| 3 | T | F | 5 | - |
| 4 | F | F | 4 | + |
| 5 | F | T | 7 | - |
| 6 | F | T | 3 | - |
| 7 | F | F | 8 | - |
| 8 | T | F | 7 | + |
| 9 | F | T | 5 | - |

(a) **[1 Mark]** What is the entropy of this collection of dataset with respect to the class?

**Solution :** Since there are 4 positive and 5 negative instances,

Entropy = - [(4/9) log (4/9) + (5/9) log (5/9)] = 0.9911

(b) **[4 Marks]** Find the Information gain of a1 and a2 and hence evaluate which amongst a1 and a2 is the better split at the root node level? Assume entropy is used as the node impurity measure.

**Solution :** Entropy before split = Entropy calculated using the full data = 0.9911 (calculated in part a)).

| | **Split performed on attribute a1** | | **Split performed on attribute a2** | |
|---|---|---|---|---|
| **Split criterion** | a1 = T :<br>( 3 + and 1 -) | a1 = F :<br>( 1 + and 4 -) | a2 = T :<br>( 2 + and 3 -) | a2 = F :<br>( 2 + and 2 -) |
| **Node entropy** | 0.8113 | 0.7219 | 0.9710 | 1 |

Weighted entropy after split based on a1 = 4/9*0.8113 + 5/9*0.7219 = 0.762

Weighted entropy after split based on a2 = 5/9*0.9710 + 4/9*1 = 0.984

Information Gain when split is performed on a1 = 0.9911 - 0.762 = 0.2291

Information Gain when split is performed on a2 = 0.9911 - 0.984 = 0.0071

Information gain is higher for split performed on a1 and hence it is the better of the two candidate splits.

(c) **[5 Marks]** Consider attributes a2 and a3 in the table above. Assuming Gini index is used as the node impurity measure evaluate which of the following is the better candidate split?

- Attribute a2 : split based on category values
- Attribute a3 : split based on threshold = mean value of a3 computed from the data in table above.

Recall that if $p_1,...,p_K$ are the estimated probabilities of instances belonging to classes 1, 2,...,K respectively, the Gini index is given by

$$\text{Gini} = 1 - \mathbf{\Sigma}\, p_i^2$$

**Solution :**

Impurity before split = Gini index calculated using the full data

$$= 1 - [(4/9)^2 + (5/9)^2] = 0.4938$$

| | **Split performed on attribute a2** | | **Split performed on attribute a3** | |
|---|---|---|---|---|
| **Split criterion** | a2 = T :<br>( 2 + and 3 -) | a2 = F :<br>( 2 + and 2 -) | a3 ≤ 46 / 9<br>( 2 + and 3 -) | a3 > 46 / 9<br>( 2 + and 2 -) |
| **Node impurity** | 0.48 | 0.50 | 0.48 | 0.50 |

Weighted impurity after split based on a2 = 5/9*0.48 + 4/9*0.50 = 0.489

Weighted impurity after split based on a3 = 5/9*0.48 + 4/9*0.50 = 0.489

Information Gain when split is performed on a2 = 0.4938 - 0.489 = 0.0048

Information Gain when split is performed on a3 = 0.4938 - 0.489 = 0.0048

Information gain is the same for both candidate splits.